Imperial College London

Lecture 7

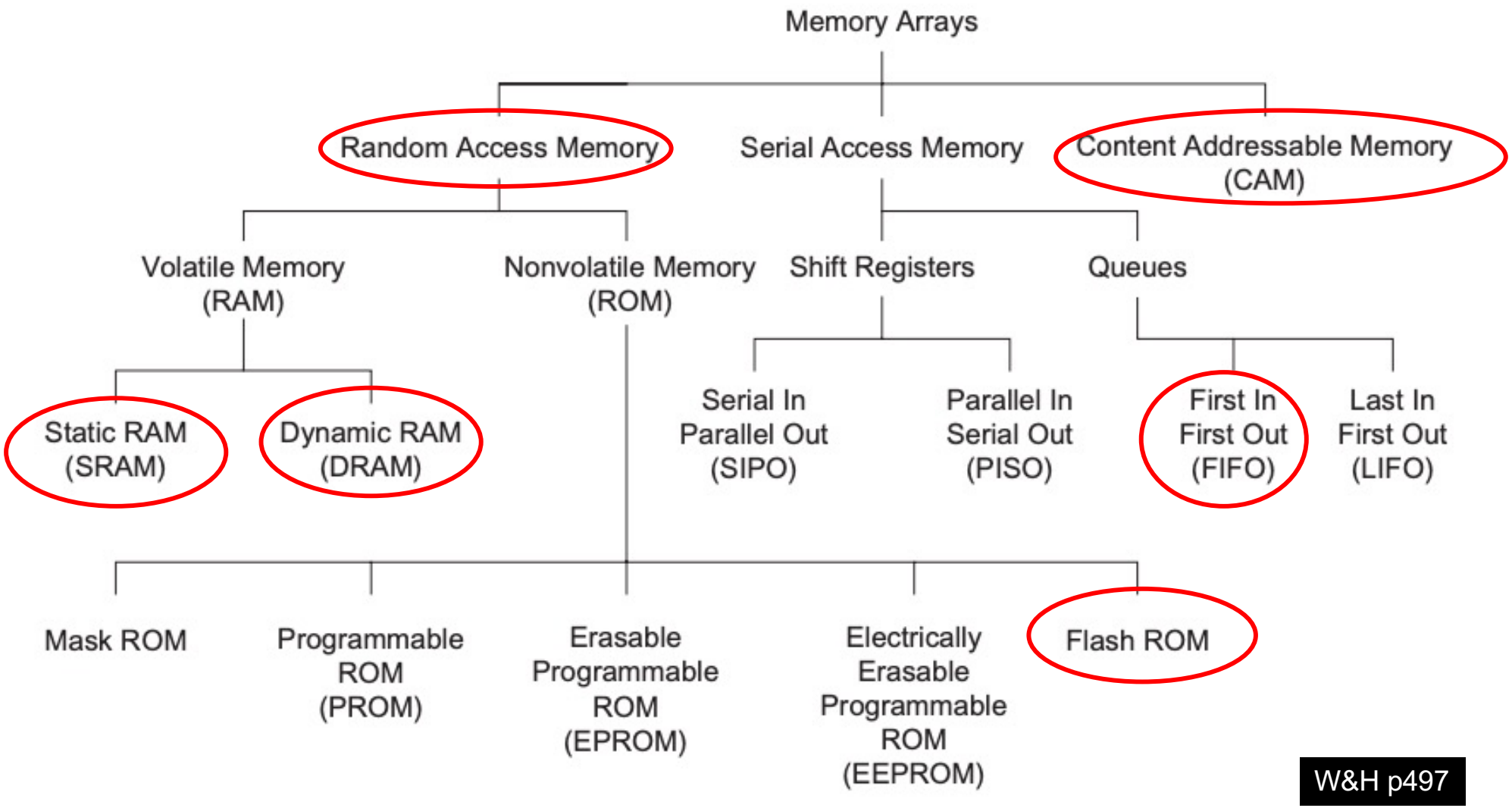
Array Subsystems

Prof Peter YK Cheung

Department of Electrical & Electronic Engineering

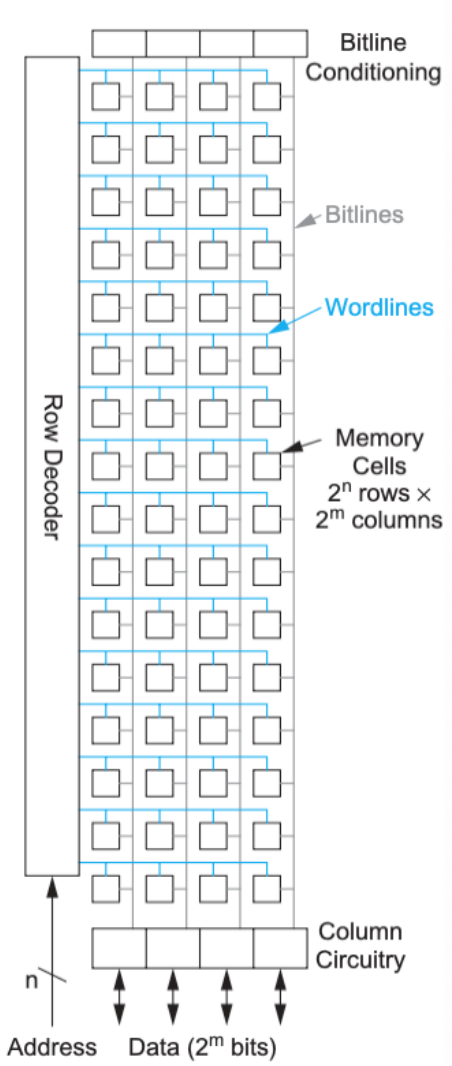
URL: https://elec70142-digital-vlsi-design.github.io/course-website/
E-mail: p.cheung@imperial.ac.uk

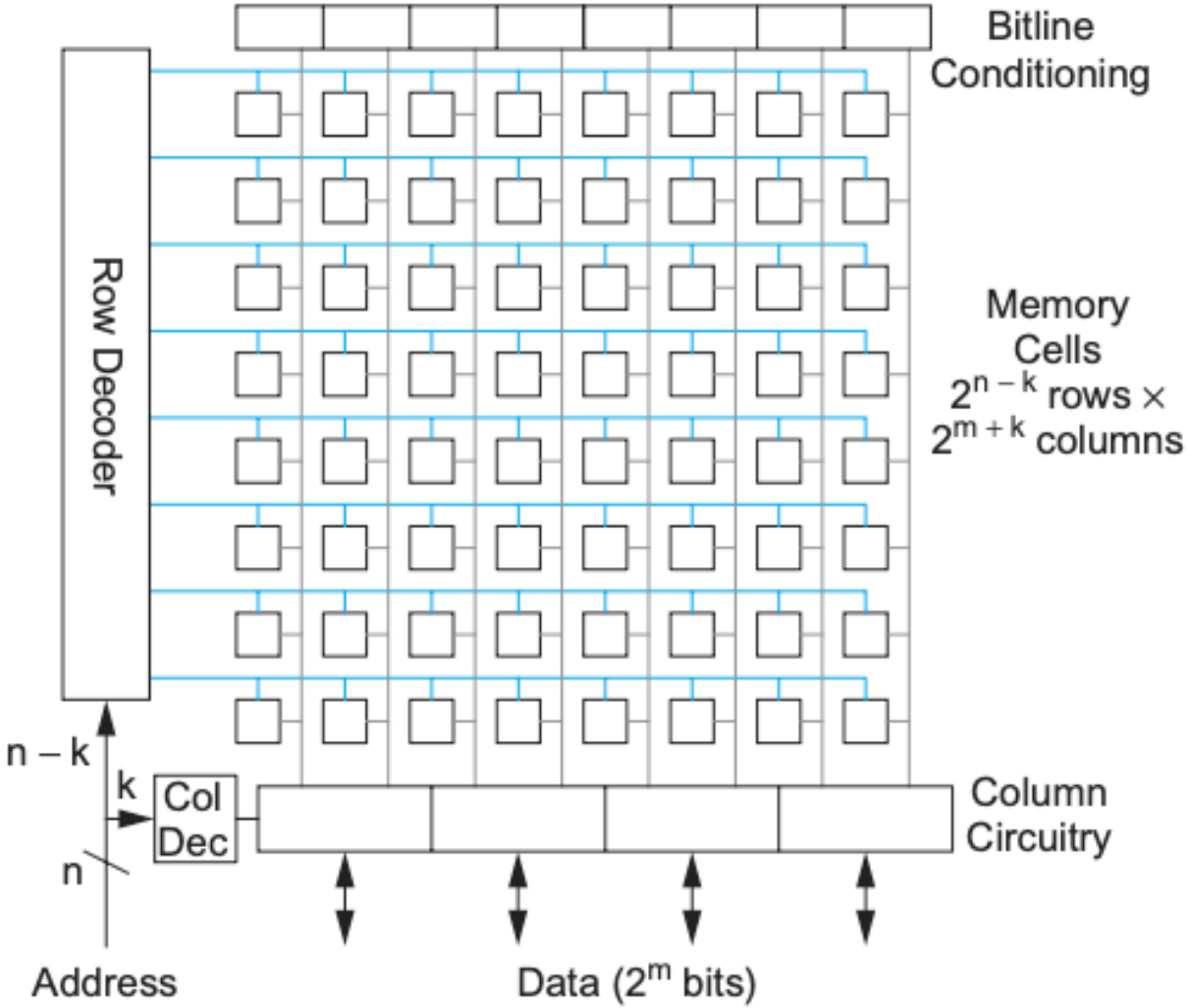
Categories of Memory Arrays



W&H p497

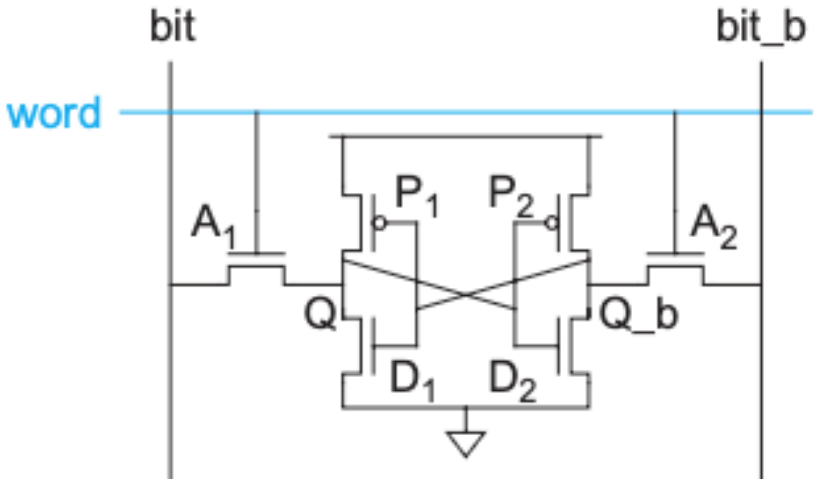
Memory Array Architecture

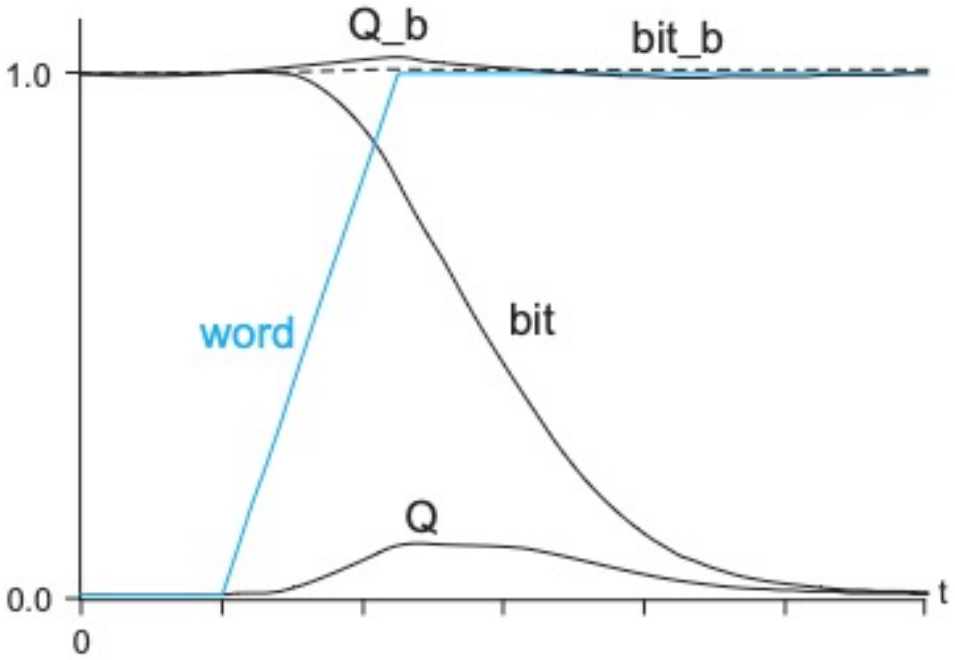




W&H p499-500

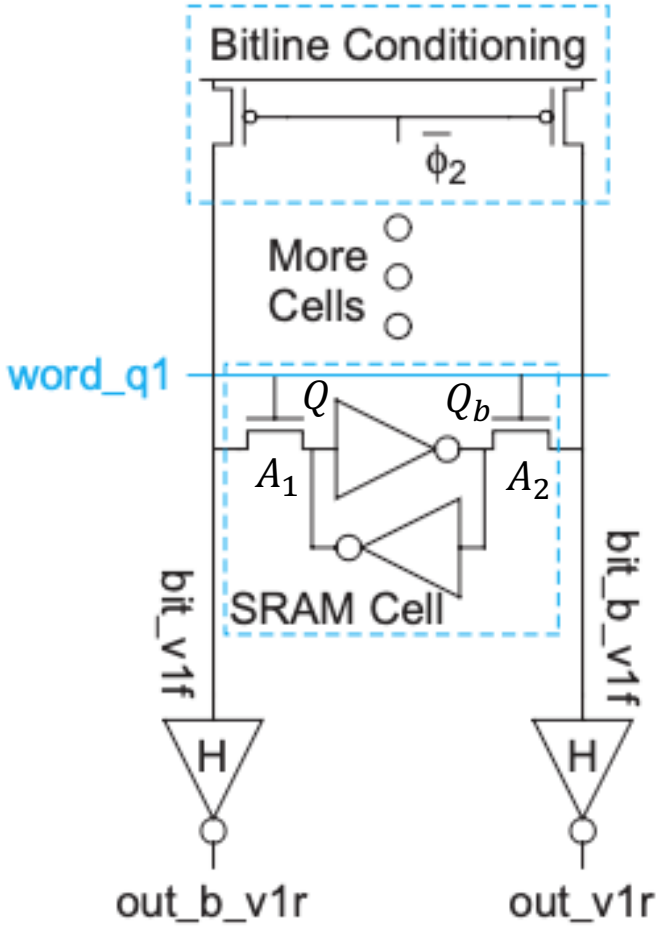
Read operation for SRAM cell

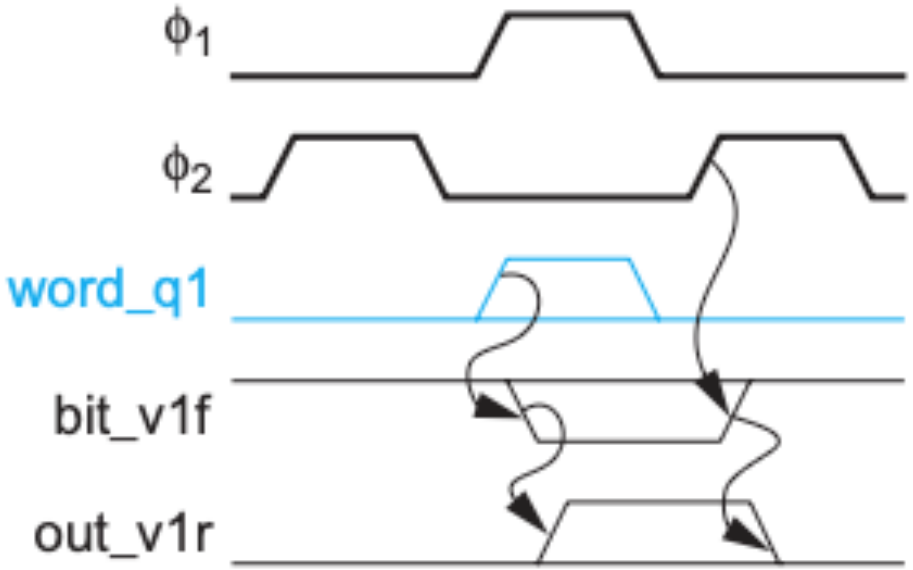




W&H p500

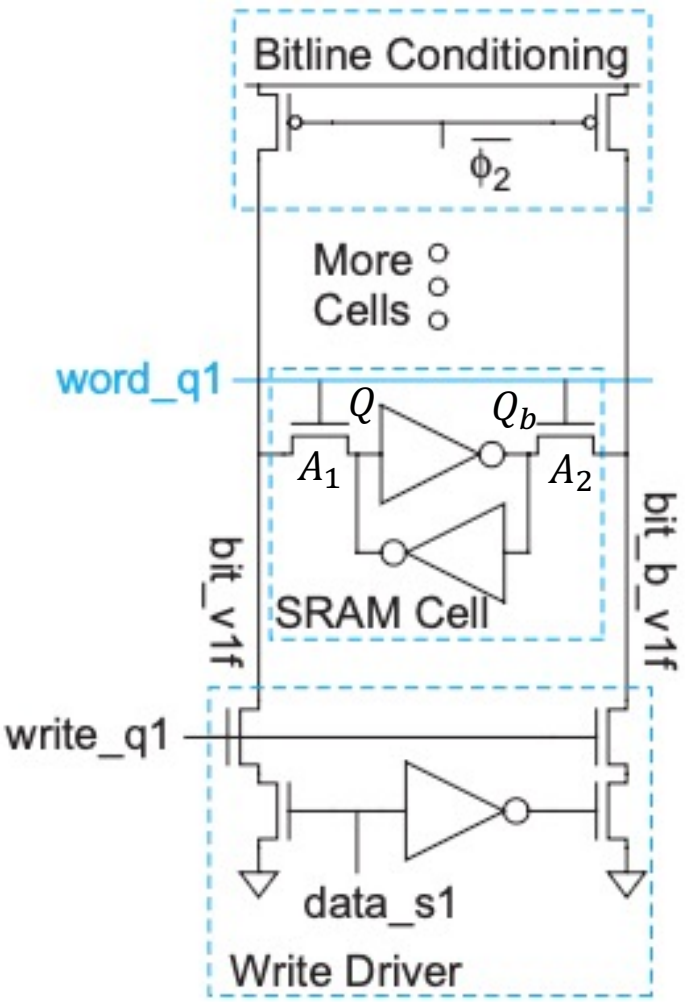
Column Read Operation

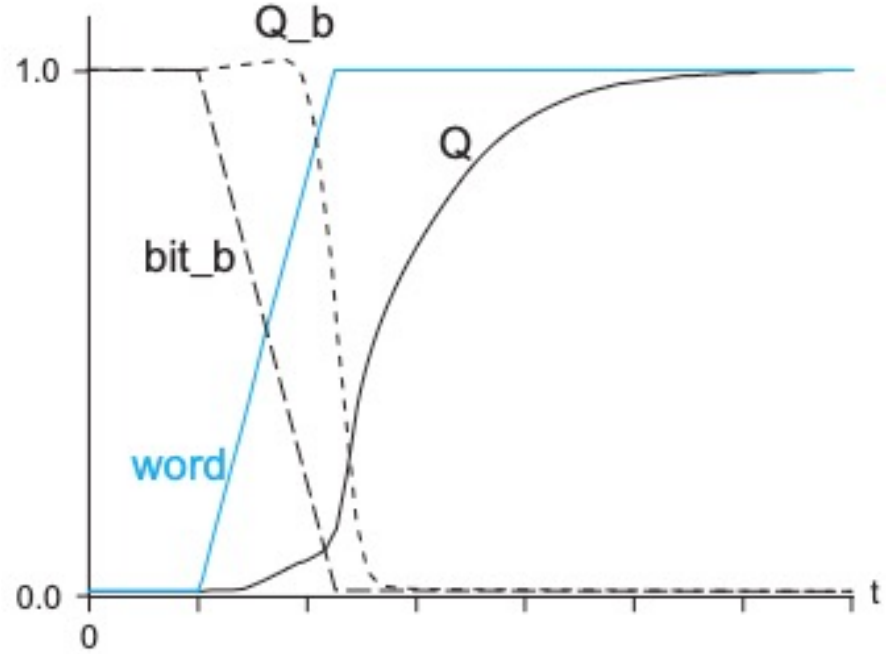


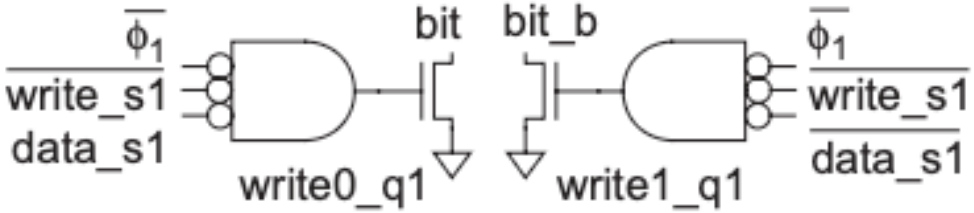


W&H p501

Write Operation

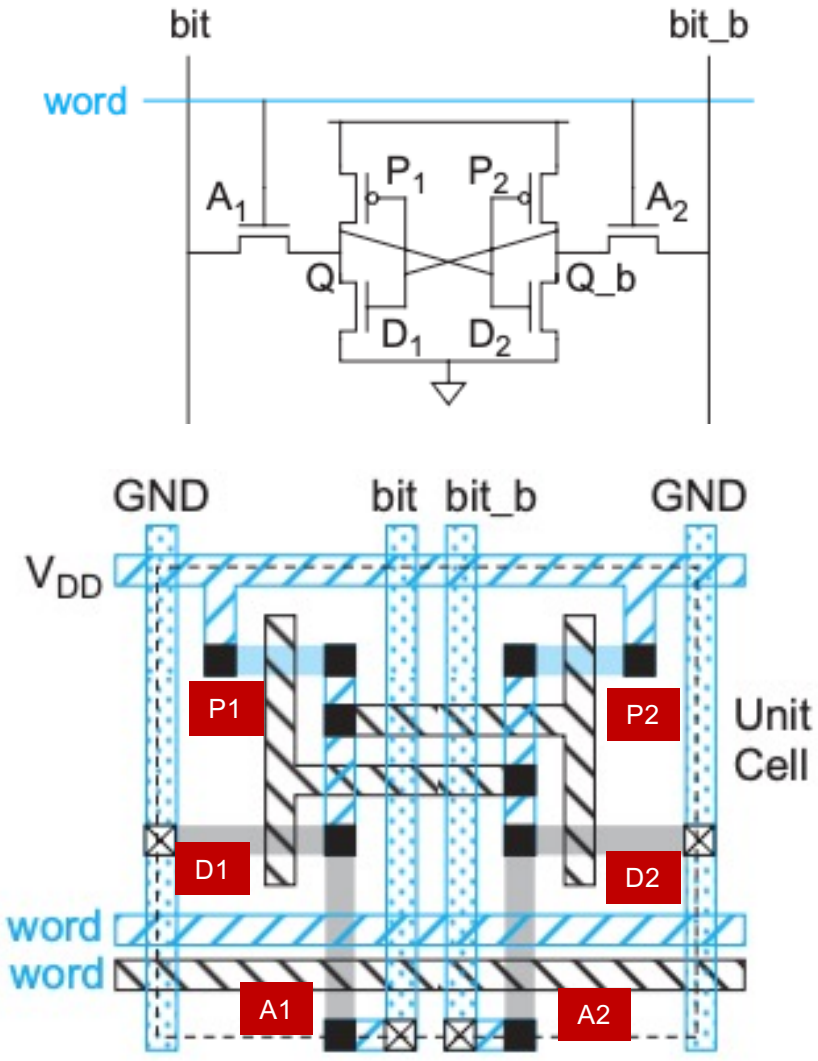


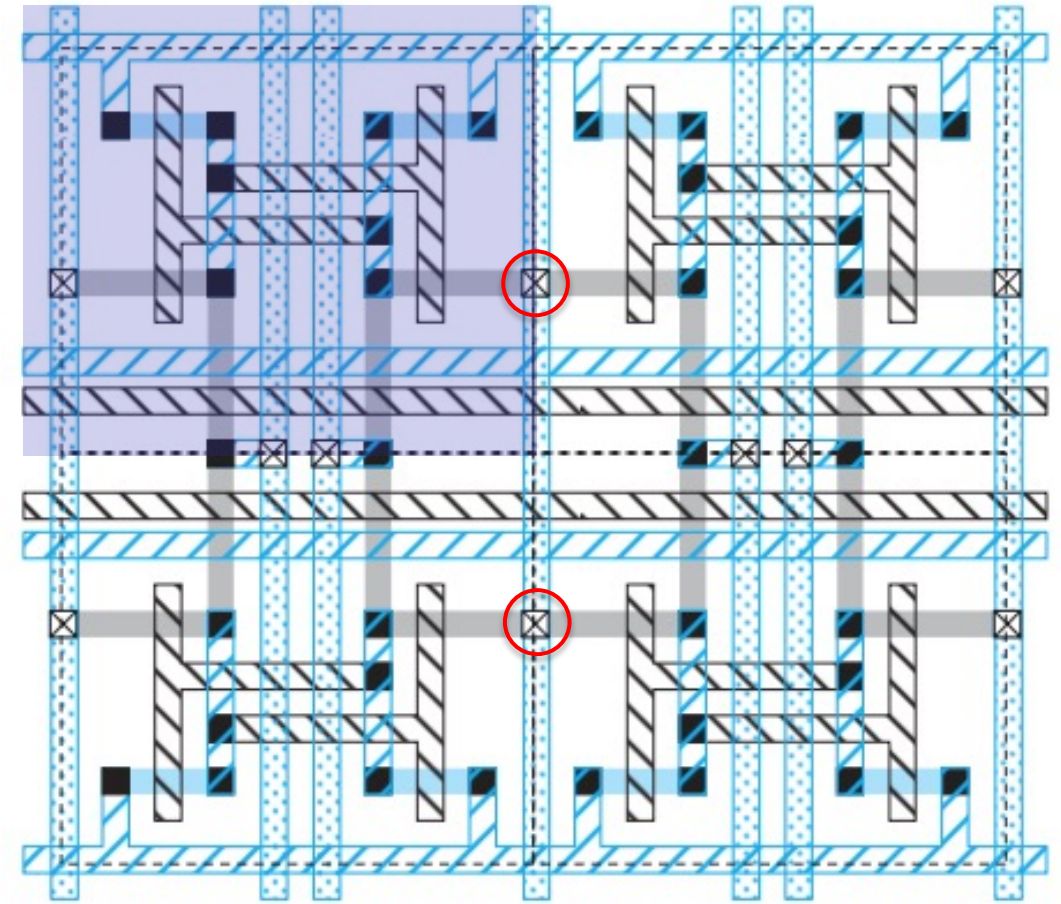




W&H p501

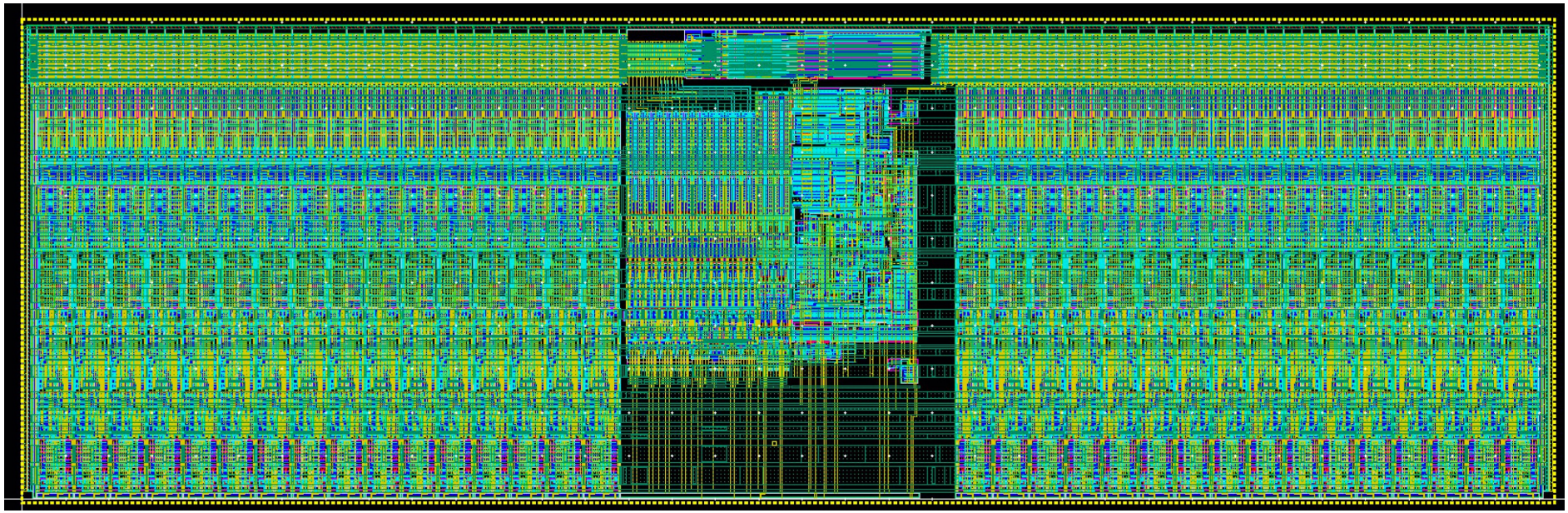
6T SRAM cell Layout (stick diagram)



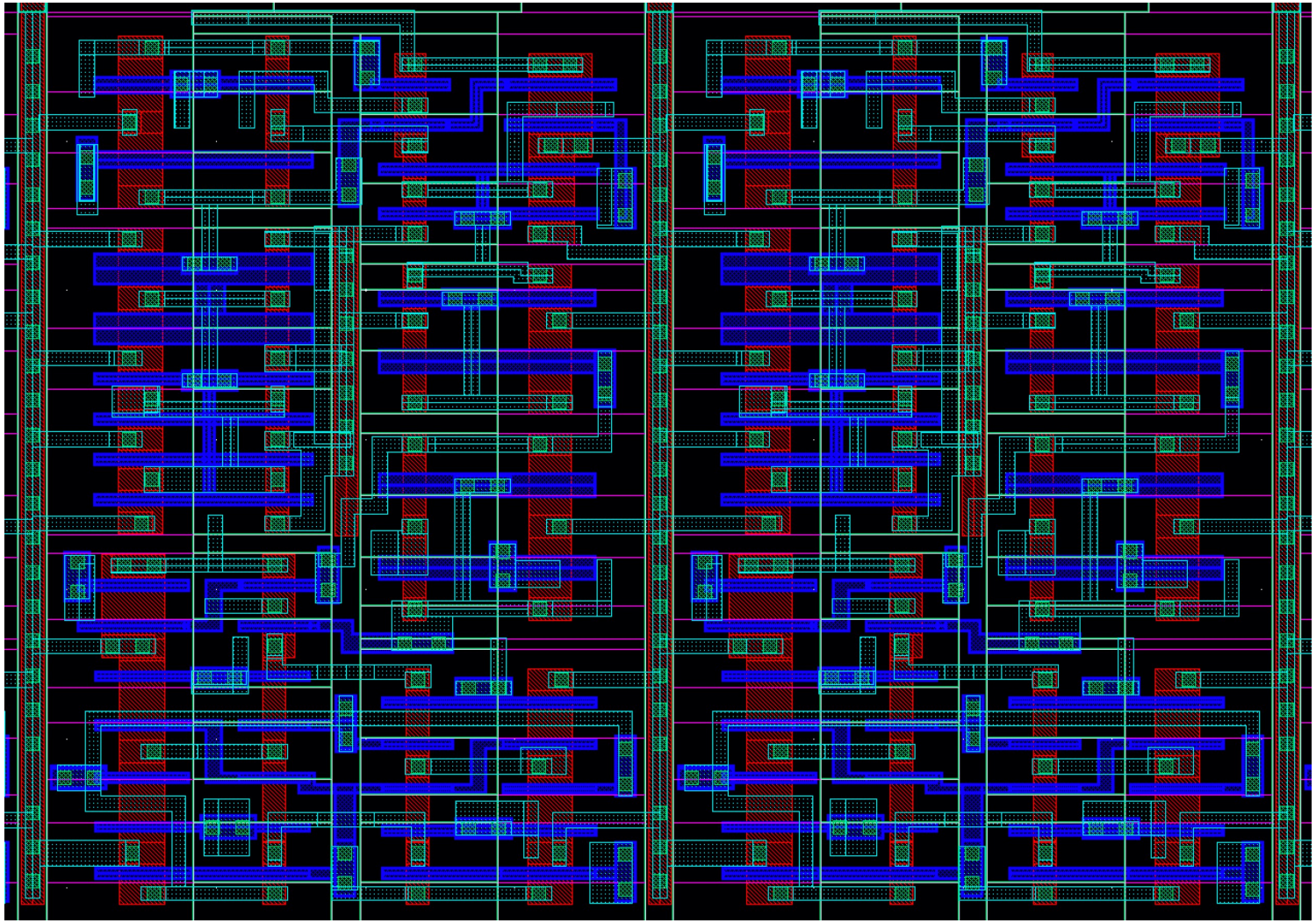


W&H p504

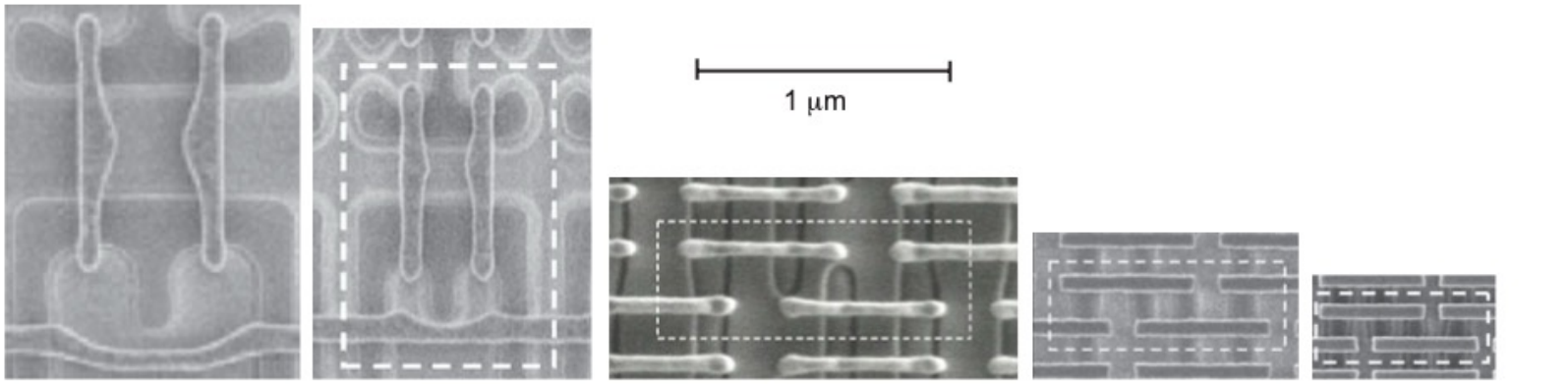
32 x 32 SRAM

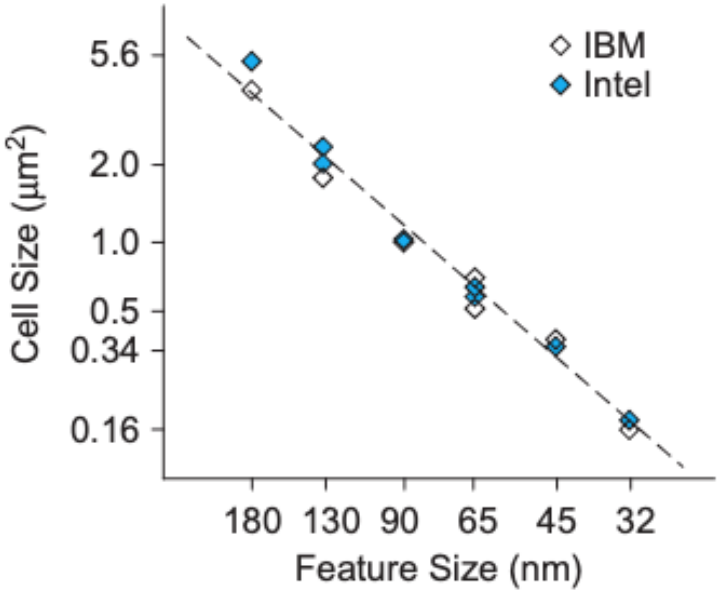


Zoom-in RAM cells



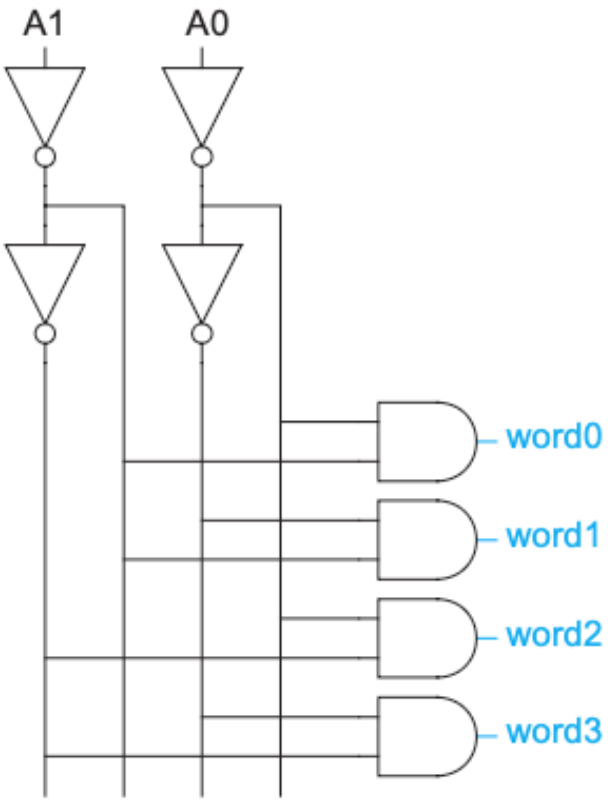
SRAM Scaling

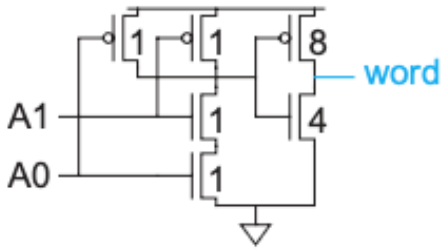


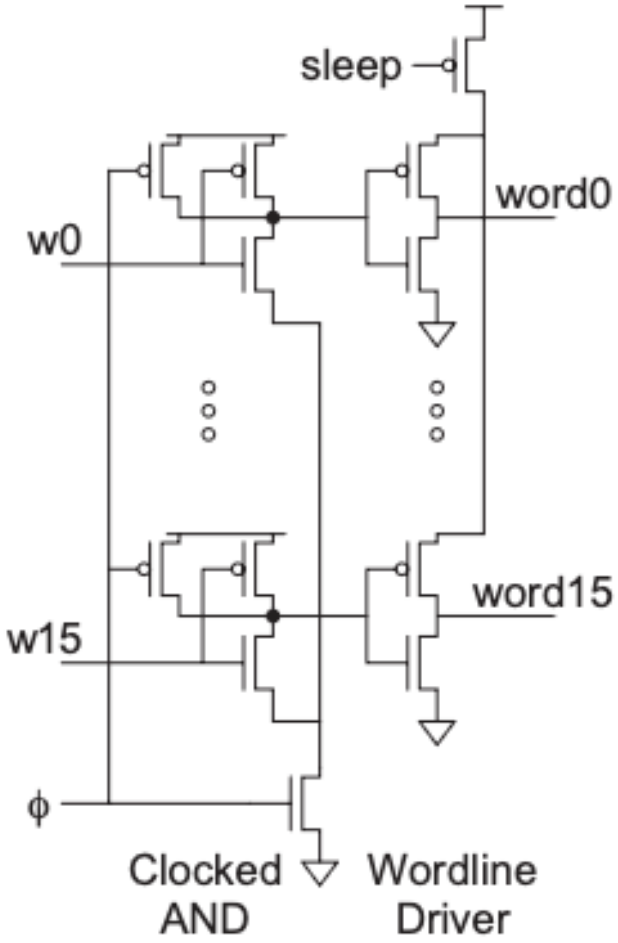


W&H p505

Row Decoder Circuits



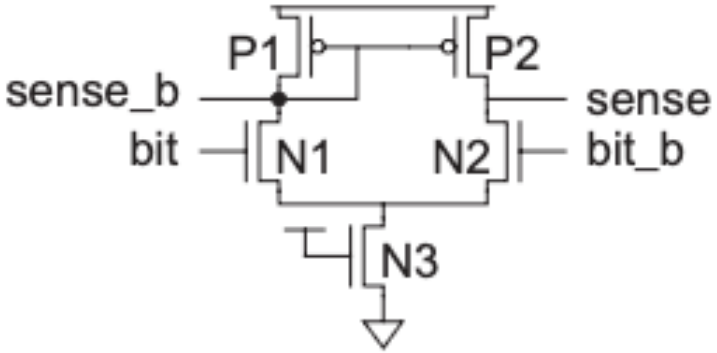




W&H p506-507

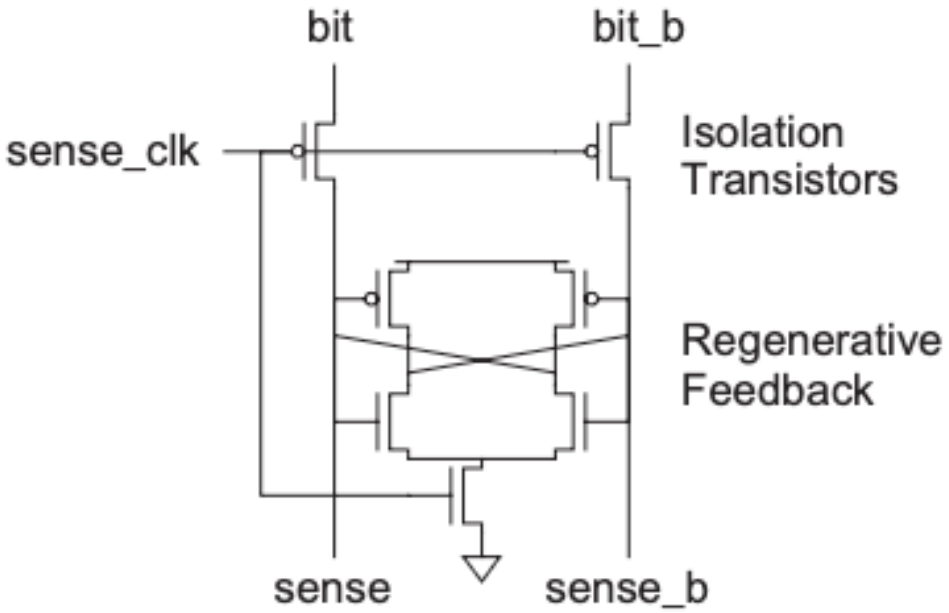
Sense Amplifier

Differntial Sense Amplifier



(a)

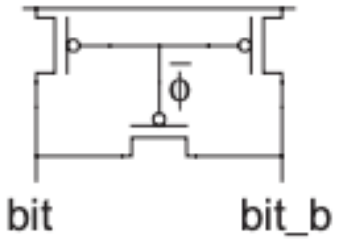
Clocked Sense Amplifier



(b)

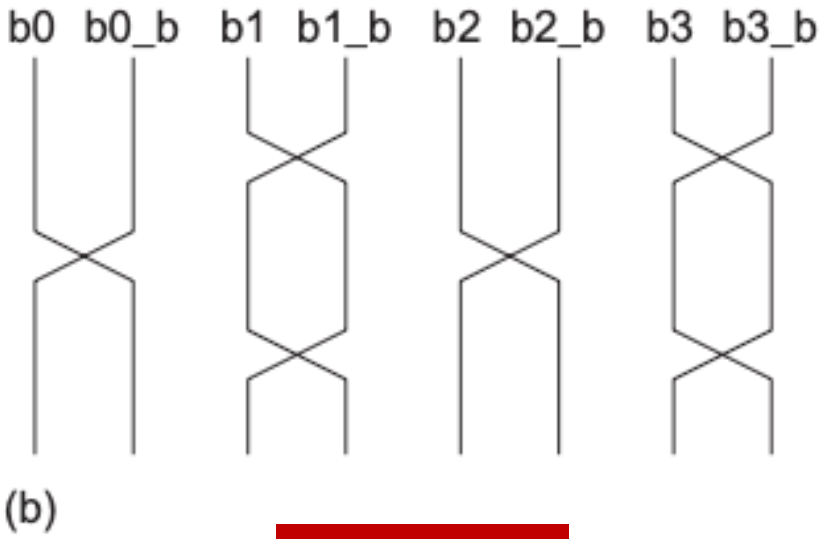
W&H p511

Differential Noise Reduction



(a)

Bitlines equalization

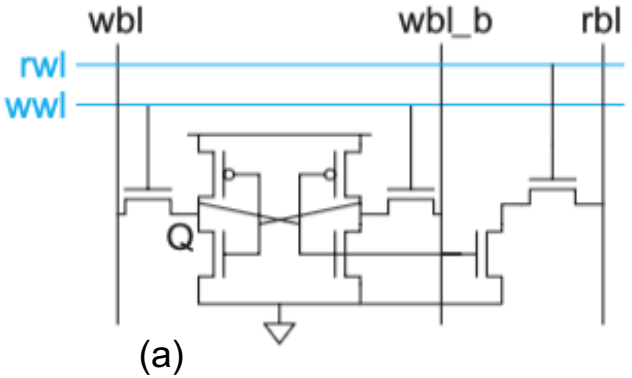


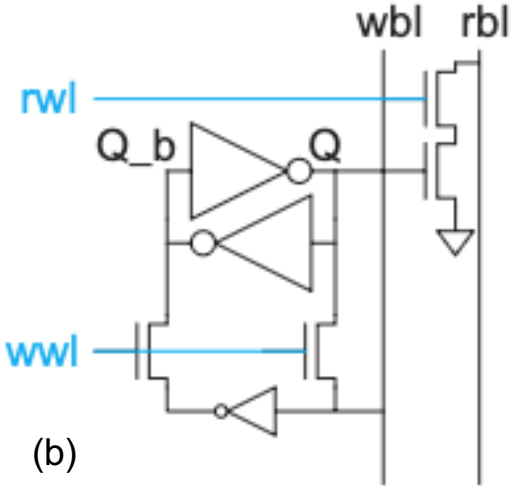
(b)

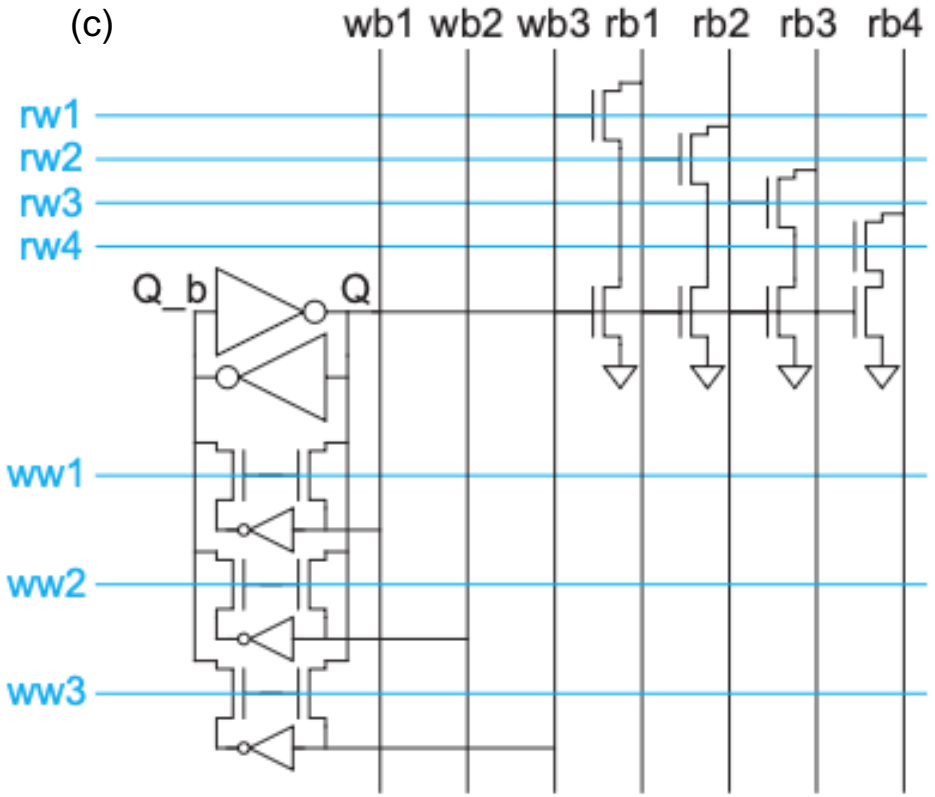
Bitlines twisting

W&H p513

Multiport SRAM



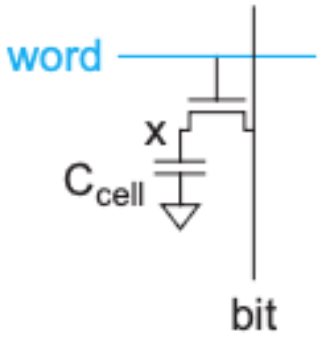


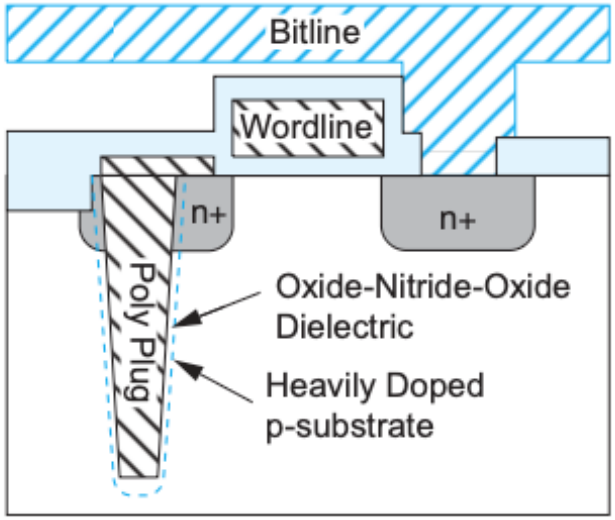


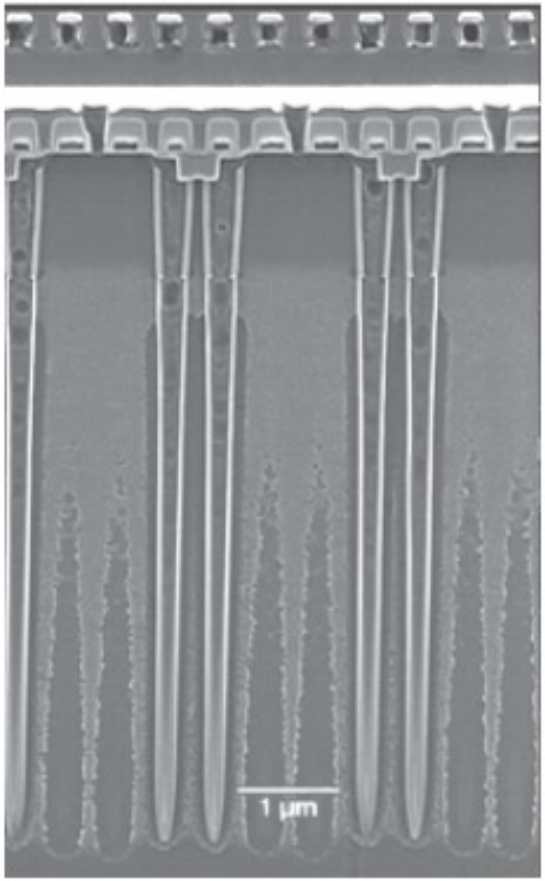
W&H p514-515

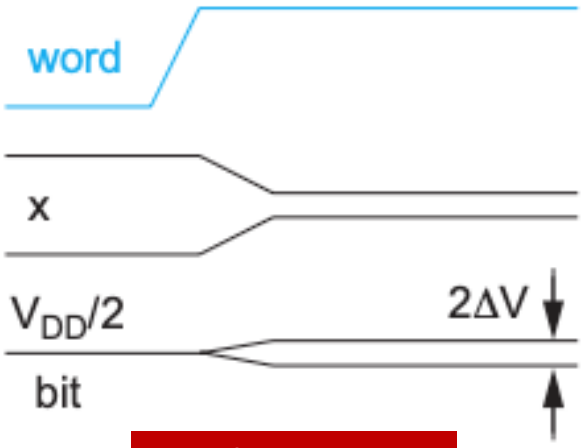
DRAM cell

Trench capacitor









VDD/2 precharge

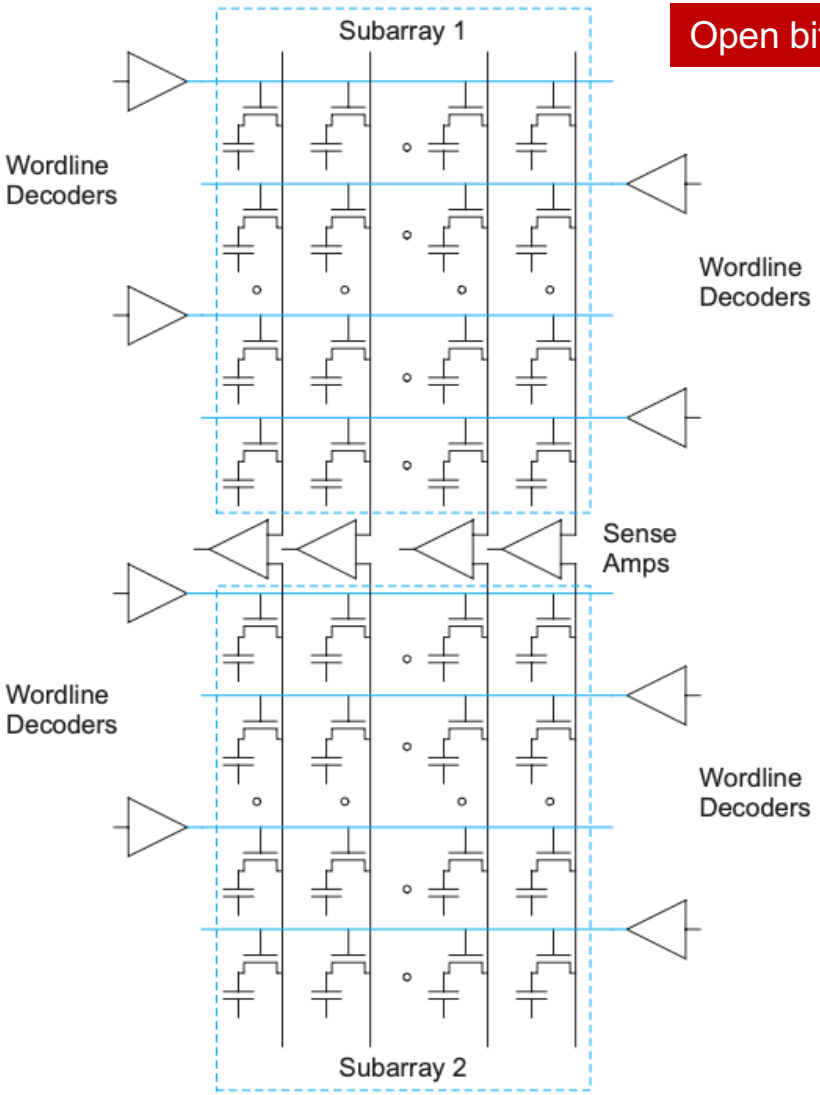
$$\Delta V = \frac{V_{DD}}{2}\frac{C_{cell}}{C_{cell}+C_{bit}}$$

Charge sharing

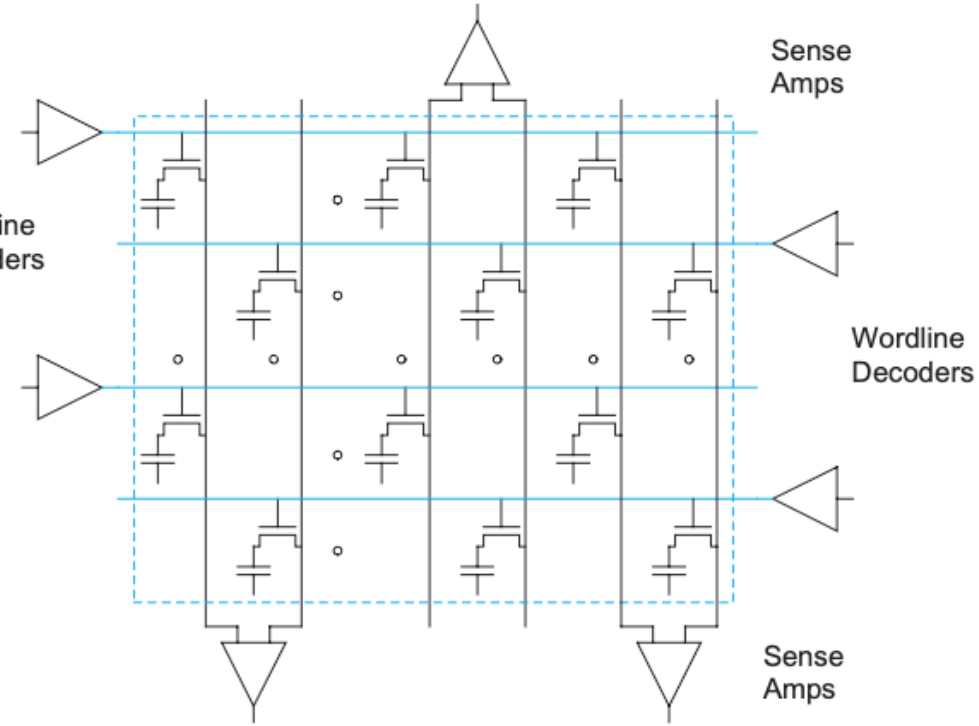
W&H p522-523

DRAM array

Open bitline



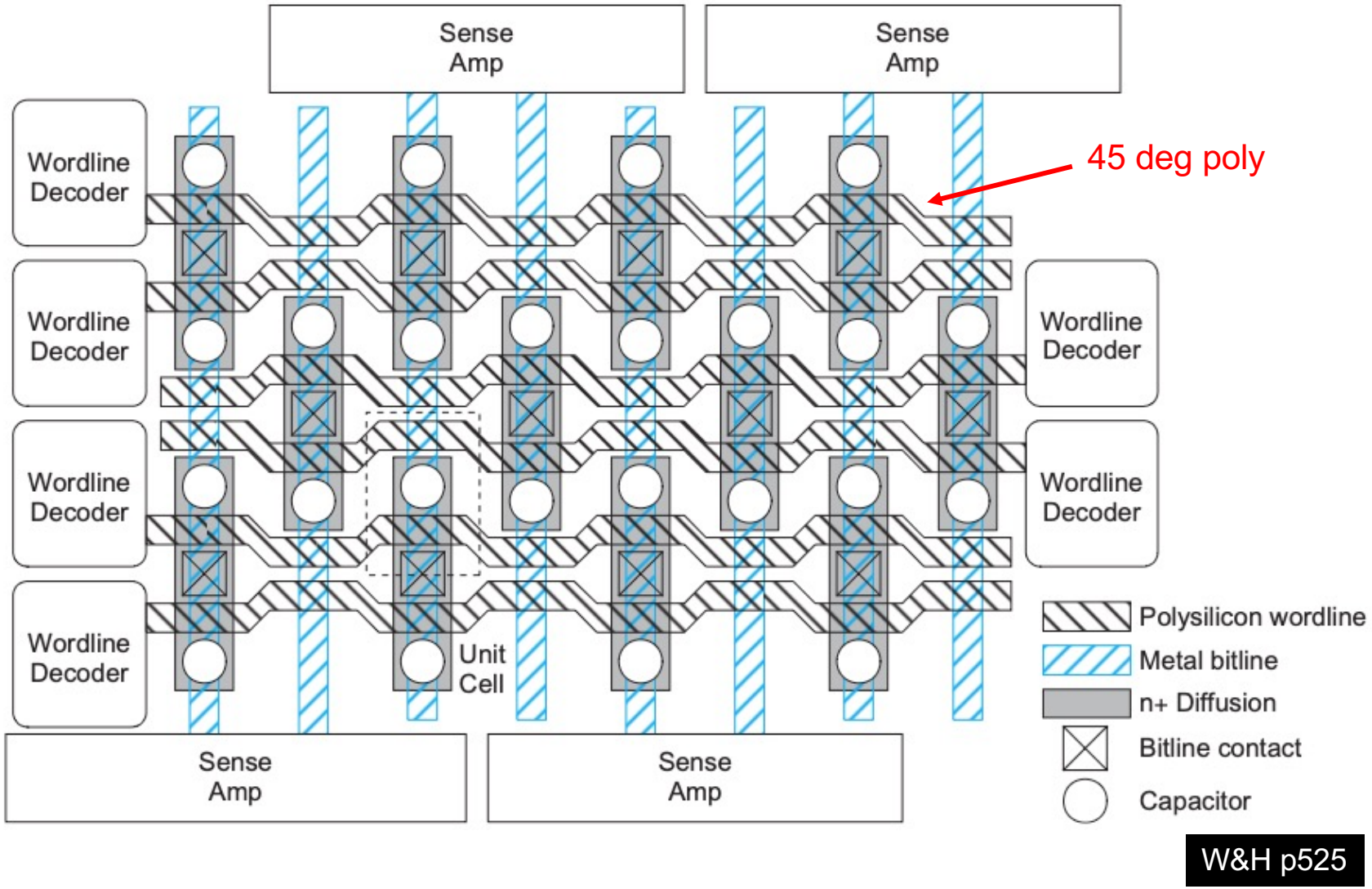
Folded bitline



- Neighbouring bitline provide reference.
- Reduce noise sensitibity

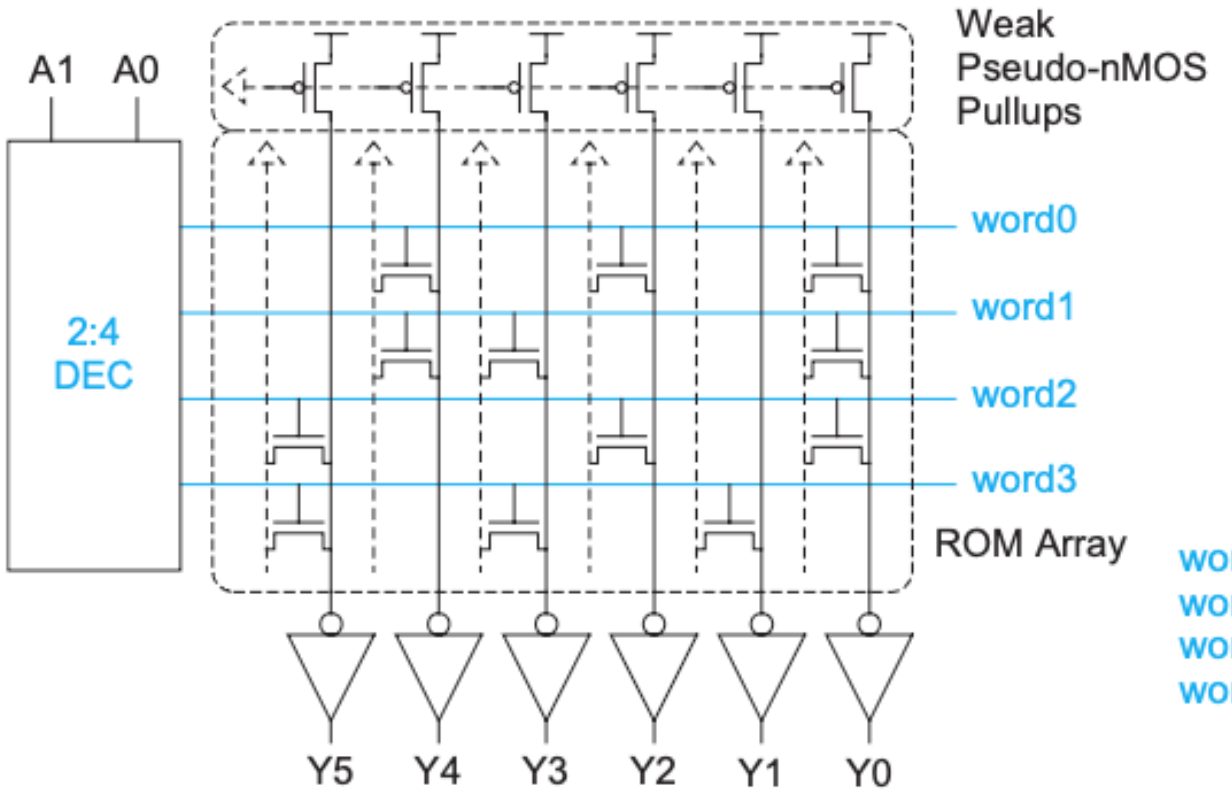
W&H p524

Folded Bitline Layout



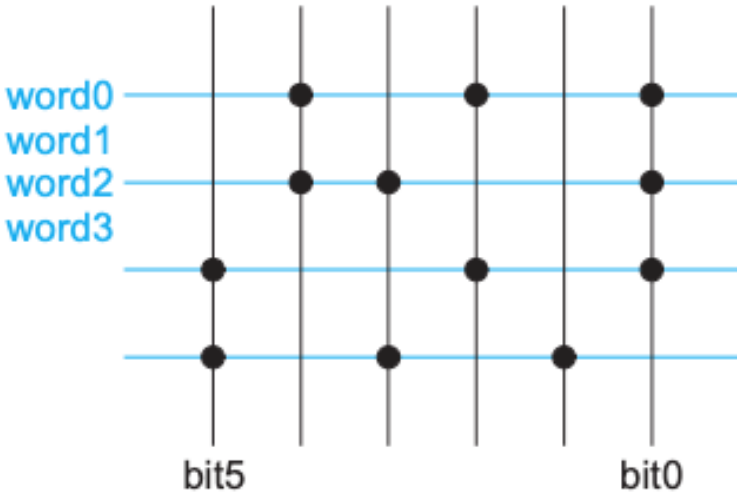
W&H p525

Pseudo nMOS ROM Circuit



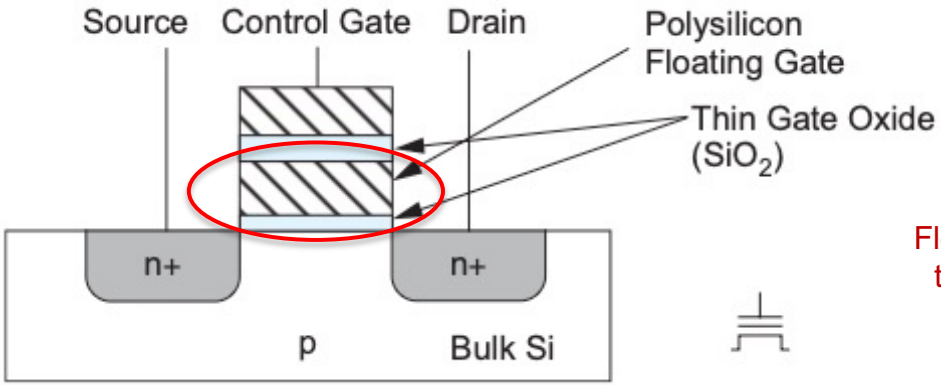
4-word by 6-bit ROM

```
word0: 010101
word1: 011001
word2: 100101
word3: 101010
```

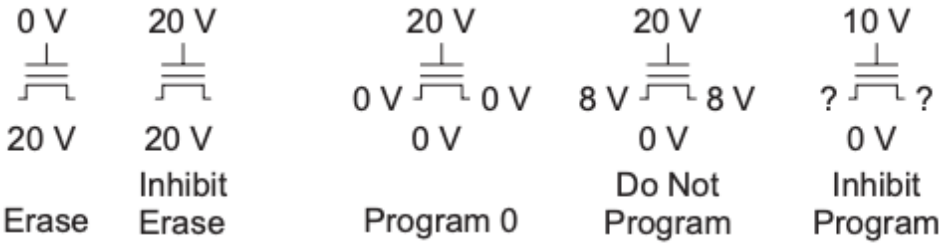


W&H p527

Flash Memory Cell



Floating gate traps electrons through Fowler-Nordheim tunnelling

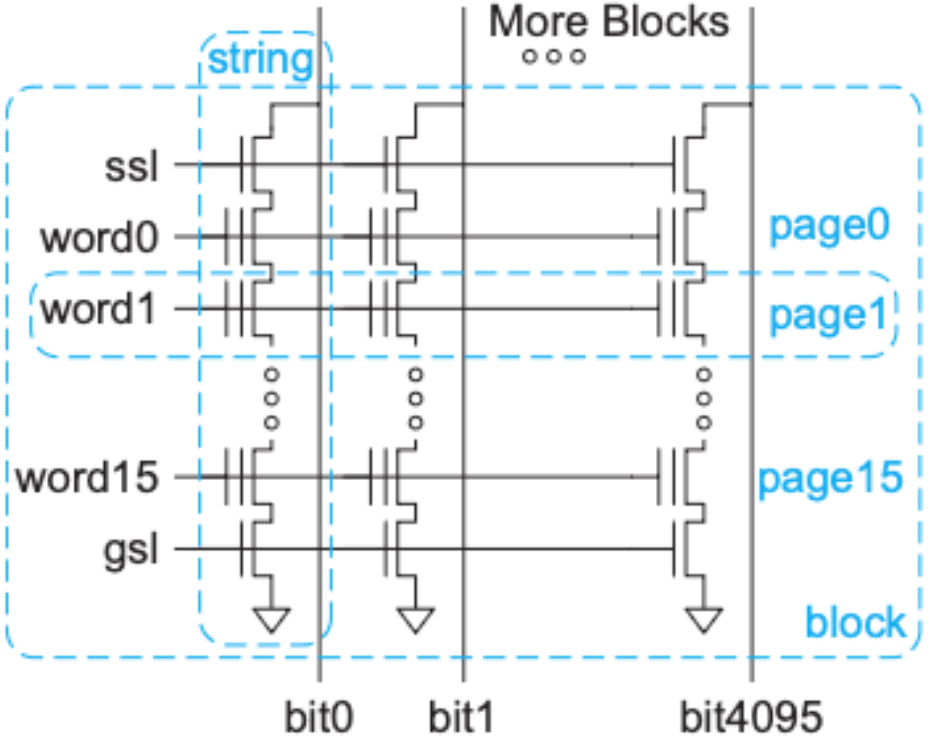


Substrate raised to 20V – cells programmed 1

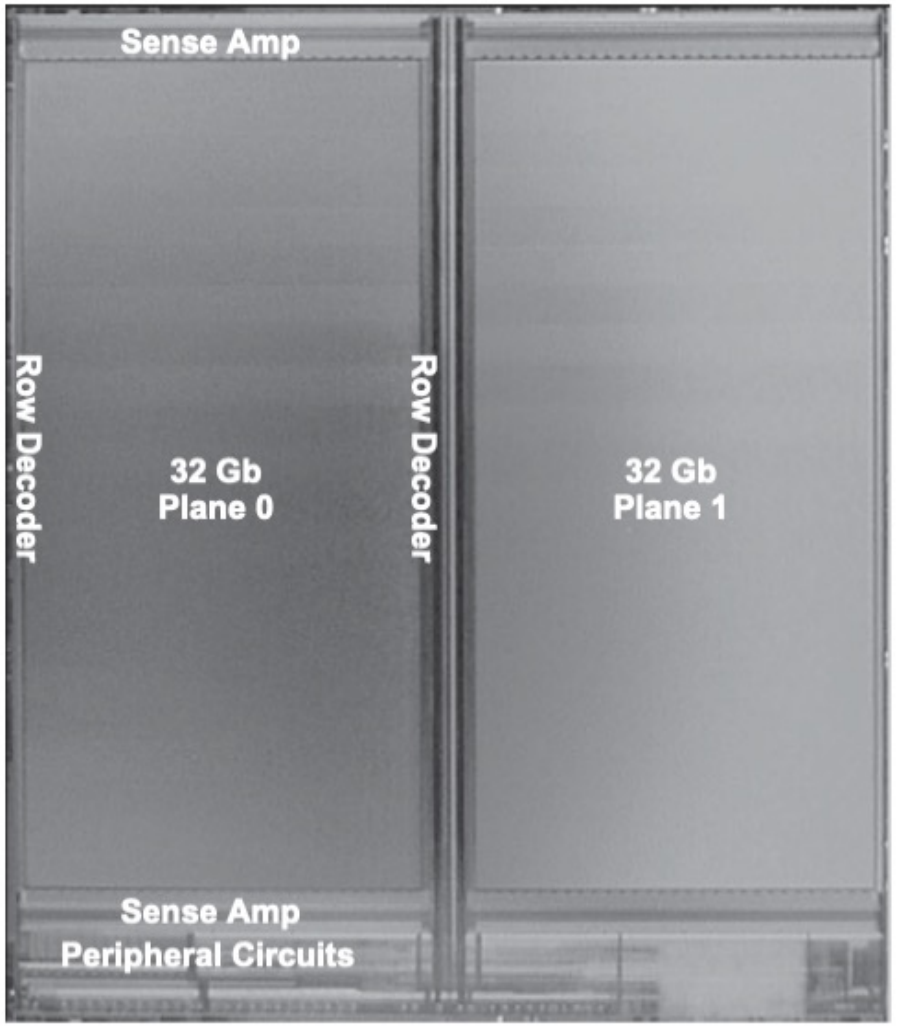
Gate raised to 20V - trap electrons raises V_t

W&H p531

Flash Memory Array

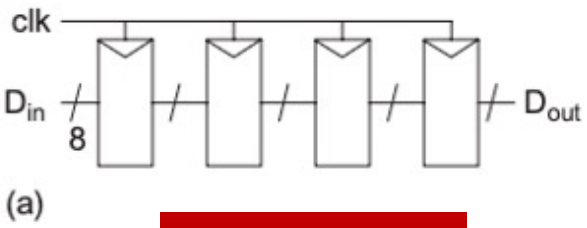


Each cell can store multibit value!

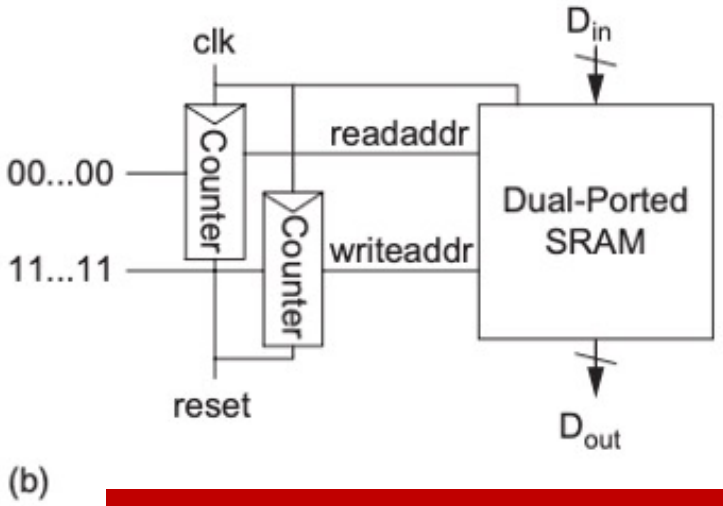


W&H p531

Shift Registers

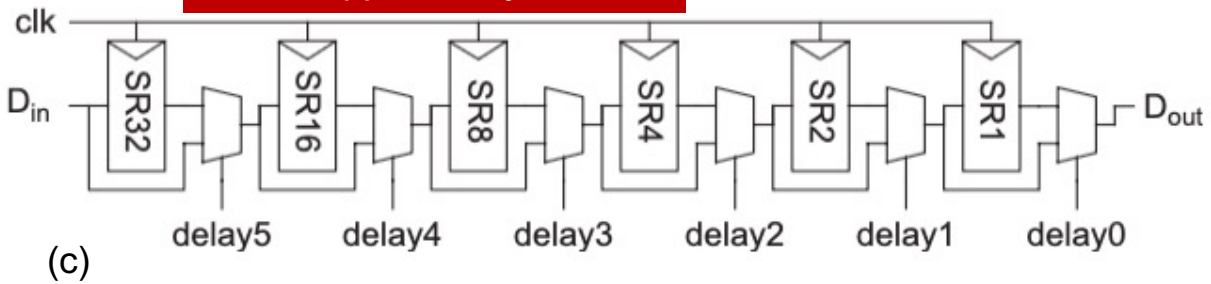


8-bit 4-stage SR



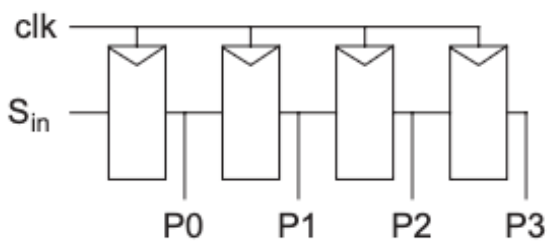
SR implemented with SRAM and counters

SR as tapped delay for DSP

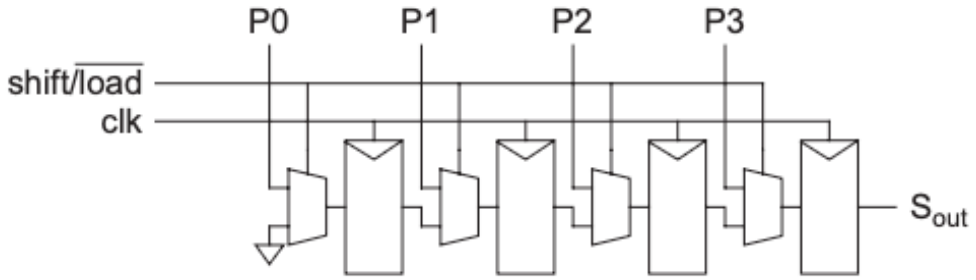


W&H p534

Delay Line and FIFO

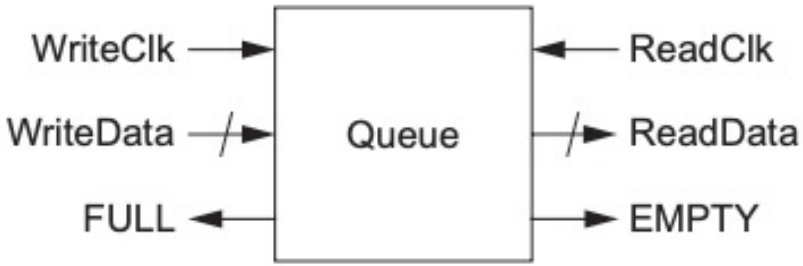


SIPO memory



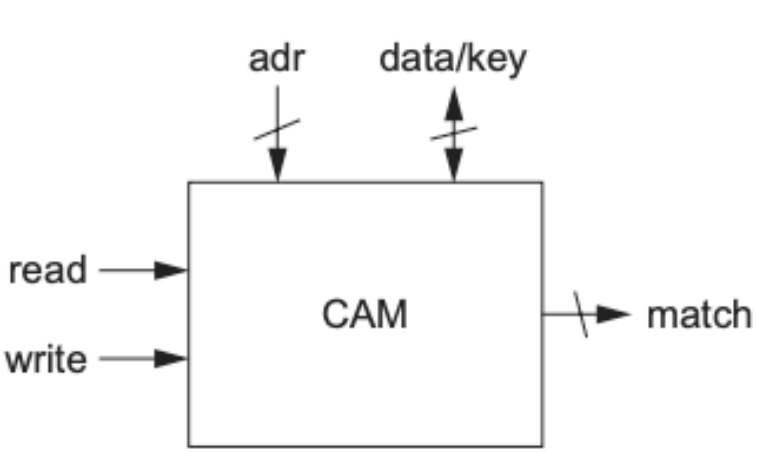
PISO memory

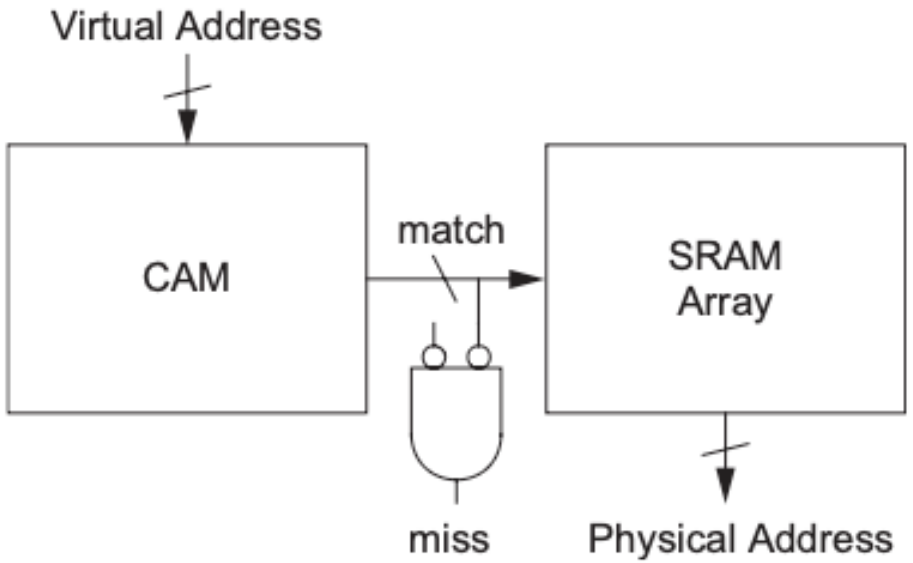
FIFO buffer



W&H p534

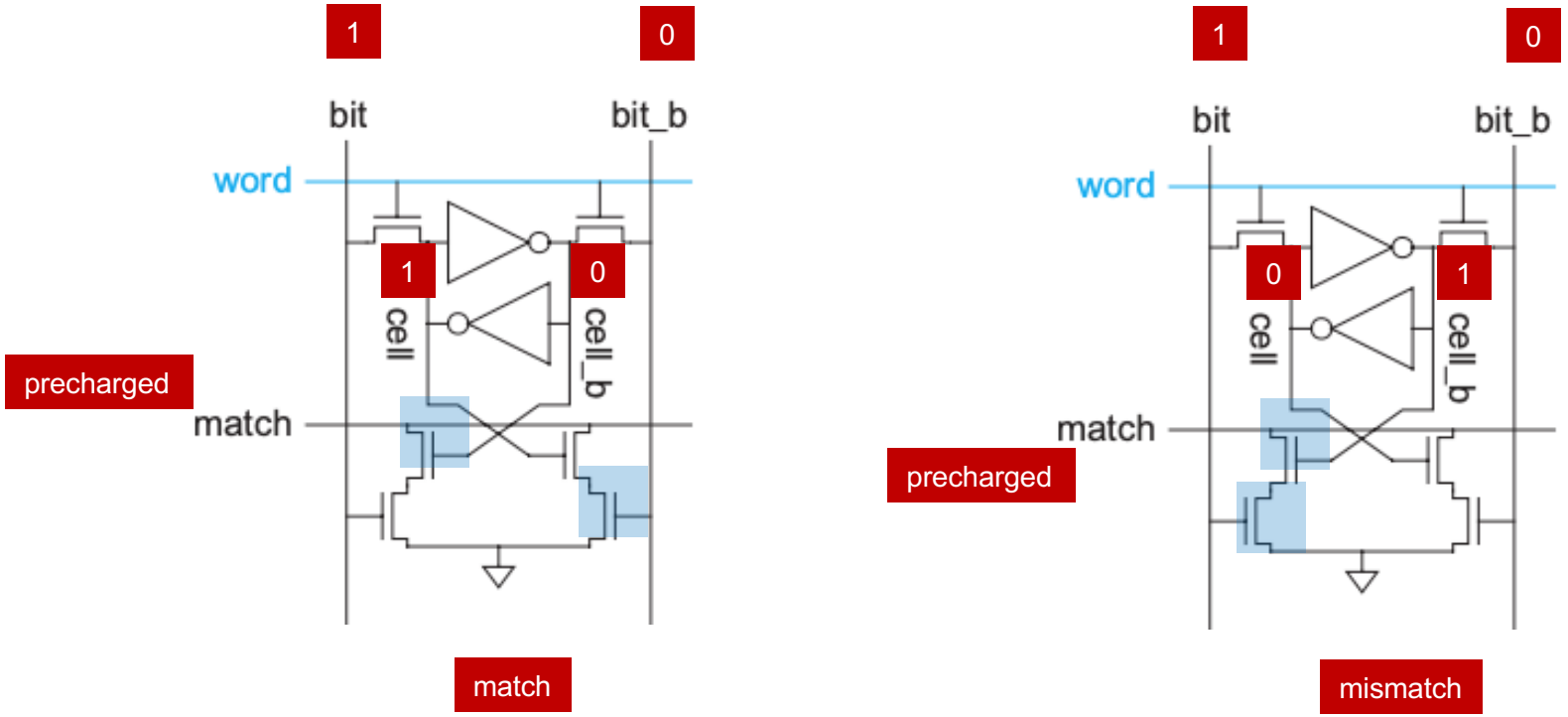
Content Addressable Memory (CAM)





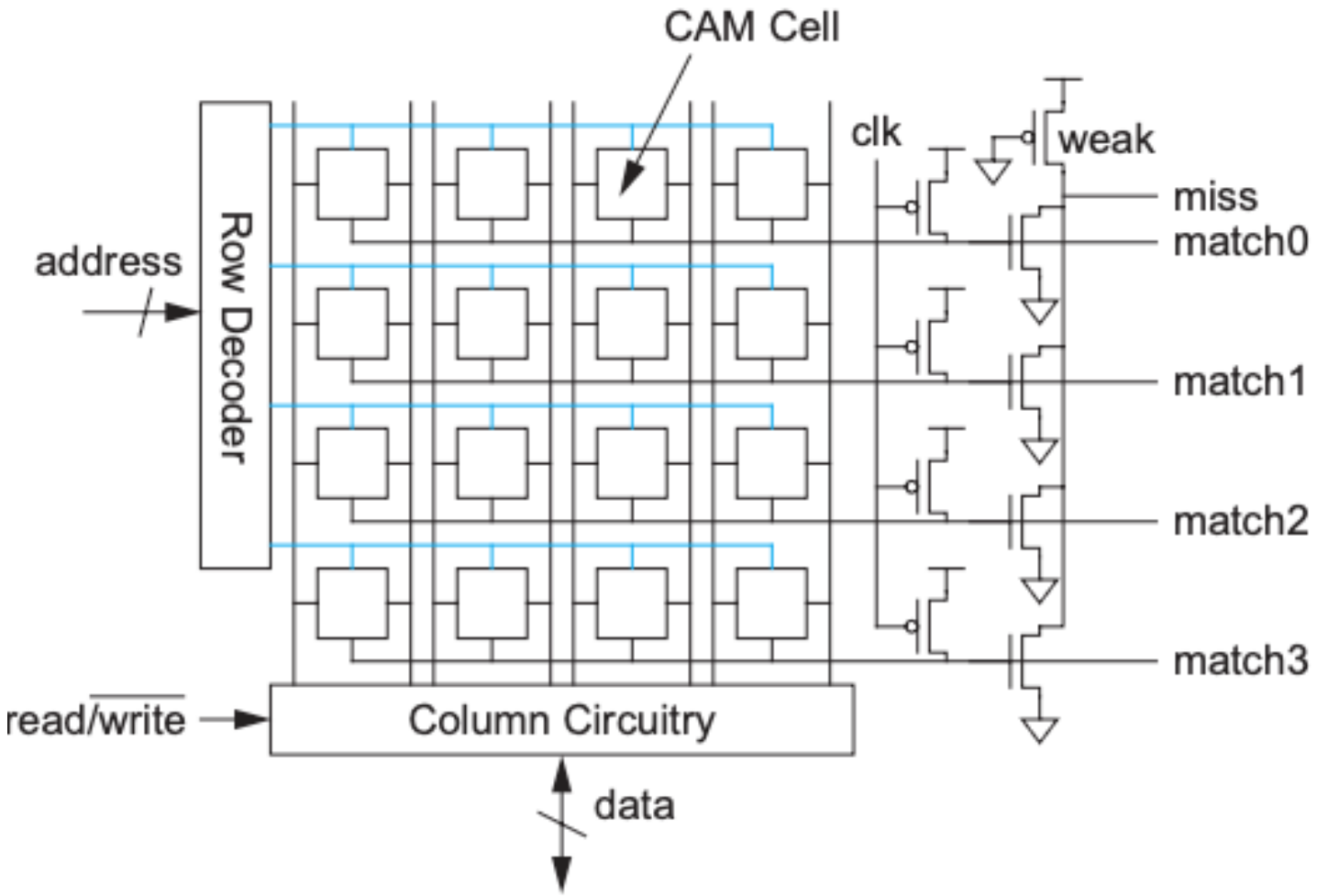
W&H p535

CAM Circuit (1)



W&H p535

CAM Circuit (2)

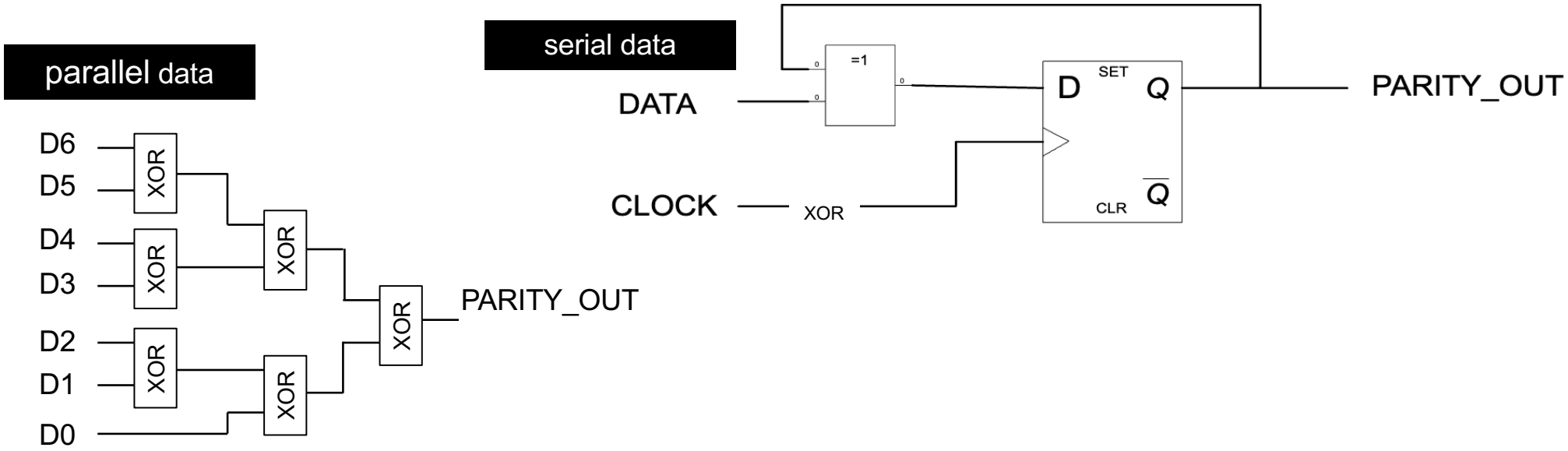


W&H p535

Error Detection in Memory

- Add extra data bit to ensure total number of 1's is EVEN – even parity
- Parity bit is the exclusive-OR of all other data bits
- If one of the received bit is wrong, then number of 1's will be odd (including parity bit) and error is detected

Data	Parity bit	number of 1's
1 1 0 1 1 1 0	1	6
0 1 0 1 1 0 1	0	4



Error Correction in Memory

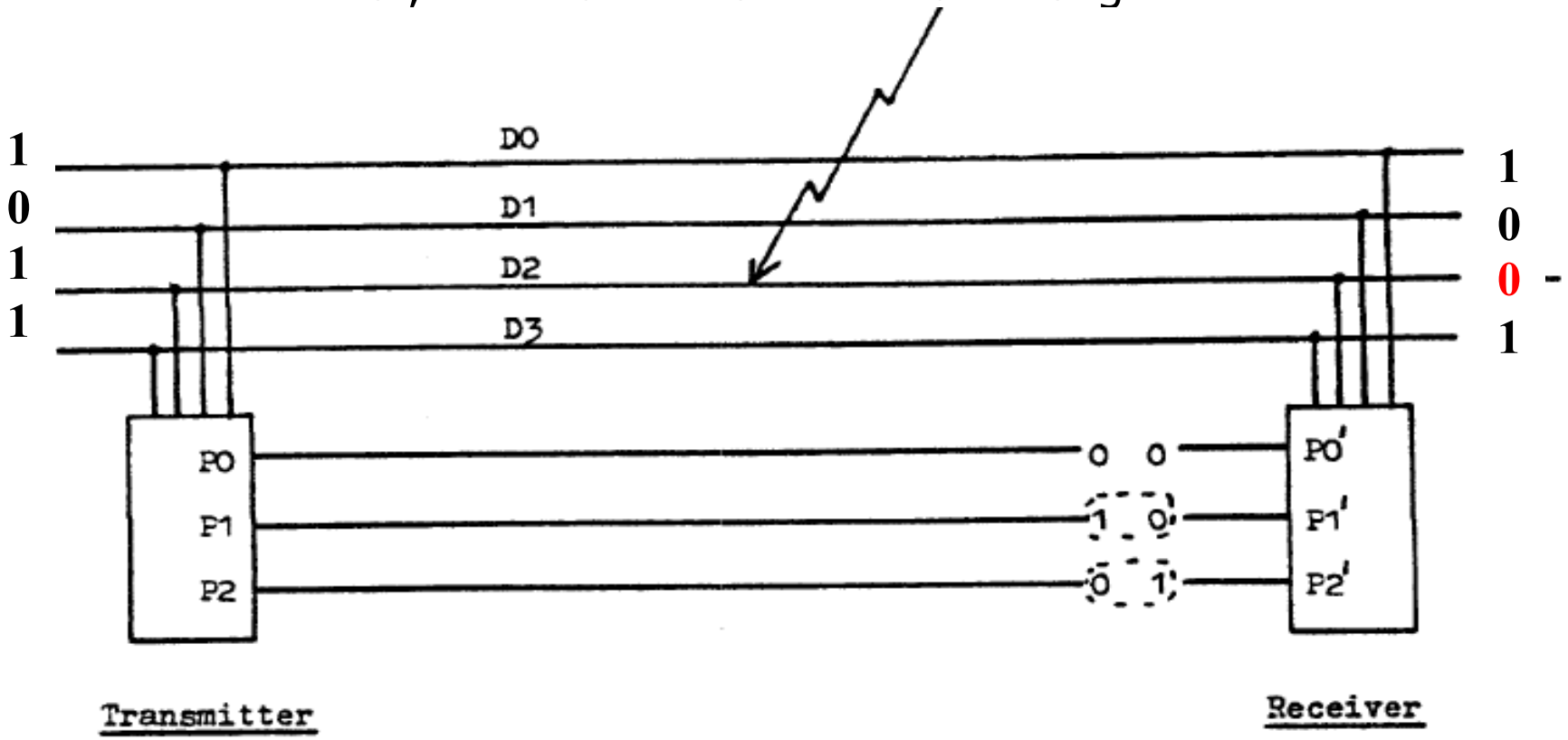
- Simple parity is limiting in its ability to detect error
- It cannot CORRECT error
- Use multiple check bits, each giving parity for a DIFFERENT COMBINATION of data bits
- If a data bit is wrong, it will affect only some check bits – from which deduce WHICH bit is wrong:

	D0	D1	D2	D3	
P0:	x	x		x	P0 = D0 ⊕ D1 ⊕ D3
P1:	x		x	x	P1 = D0 ⊕ D2 ⊖ D3
P2:		x	x	x	P2 = D1 ⊕ D2 ⊕ D3
Code:	3	5	6	7	(0,1,2,4 not used)

- **Each data bit affects a UNIQUE combination of check bits**
- **Each data bit affects at least TWO check bits – this means an error in check bit is not confused with error in data bit**

Example

- If D2 is in error, check bits P1 & P2 will be wrong:



Summary on Check Bits

- N check bits are sufficient for 2^N – N - 1 data bits
  - Each data bit must affect a different combination of check bits
  - There are 2^N possible combinations for N check bits
  - Of these, N affect only 1 bit and 1 affects no bits at all
- If you regard the combinations as binary numbers, then for 3 check bits, numbers 0, 1, 2 and 4 are ruled out, leaving 3, 5, 6, 7 as useful:

	D0	D1	D2	D3	
P0:	x	x		x	P0 = D0 ⊕ D1 ⊕ D3
P1:	x		x	x	P1 = D0 ⊕ D2 ⊖ D3
P2:		x	x	x	P2 = D1 ⊕ D2 ⊕ D3
Code:	3	5	6	7	(0,1,2,4 not used)

ECC for 8-bit data

- Detect and correct any single bit error
- Can add an extra check bit giving overall parity to both data & check bits
  - will detect double bit errors but not correct
  - For double errors, overall parity bit will be OK, but check bits is wrong

	D0	D1	D2	D3	D4	D5	D6	D7	
P0:	x	x		x	x		x		
P1:	x		x	x		x	x		
P2:		x	x	x				x	
P3:					x	x	x	x	
Code:	3	5	6	7	9	10	11	12	(0,1,2,4,8 not used, 13,14,15 spare)